

Characteristic Classes

Book Notes of *Characteristic Classes*

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1 Obstruction Theory & Characteristic Classes

1.1 A Brief Introduction to Obstruction Theory

In homotopy theory, one often encounters the following problem:

Given a pair of spaces (X, A) and a map $f : A \rightarrow Y$. Can f be extended to a map $F : X \rightarrow Y$?

Obstruction theory provides a systematic way to study this problem.

Toy Model : A map $f : S^n \rightarrow Y$ can be extended to a map $F : D^{n+1} \rightarrow Y$ if and only if f is null-homotopic.

Now we consider the general case. Let K be a CW complex, Y be a simple space, i.e., $\pi_1(Y)$ acts trivially on $\pi_n(Y)$ for all n . Under this condition, we can unambiguously use the notation $\pi_n(Y) = [S^n, Y]$. Suppose we have a map $f : K^{(n)} \rightarrow Y$, where $K^{(n)}$ is the n -skeleton of K . We want to extend f to a map $F : K^{(n+1)} \rightarrow Y$. To do this, we only need to extend f to each $(n+1)$ -cell e_α^{n+1} .

Suppose the attaching map of e_α^n is $\varphi_\alpha : S^n \rightarrow K^{(n)}$. Then $f \circ \varphi_\alpha$ is a map from S^n to Y . By the toy model, $f \circ \varphi_\alpha$ can be extended to D^{n+1} if and only if $[f \circ \varphi_\alpha] = 0$ in $\pi_n(Y)$. We in fact get a cochain $c^{n+1}(f) \in C_{cell}^{n+1}(K; \pi_n(Y))$:

$$\begin{aligned} c^{n+1}(f) : C_{n+1}^{cell}(K) &\rightarrow \pi_n(Y) \\ e_\alpha^{n+1} &\mapsto [f \circ \varphi_\alpha]. \end{aligned}$$

We call $c^{n+1}(f)$ the $(n+1)$ -**th obstruction cochain** of f . There are some basic facts about $c^{n+1}(f)$:

- f can be extended to $K^{(n+1)}$ if and only if $c^{n+1}(f) = 0$;
- If $f, f' : K^{(n)} \rightarrow Y$ are homotopic, then $c^{n+1}(f) = c^{n+1}(f')$;
- (Naturality) Suppose L is another CW complex and $\phi : L \rightarrow K$ is a cellular map, then $c^{n+1}(f \circ \phi) = \phi^\# c^{n+1}(f)$.

At the following, we give some important properties of $c^{n+1}(f)$ without proof.

Theorem 1.1. $c^{n+1}(f)$ is a cocycle, i.e., $\delta c^{n+1}(f) = 0$. Thus it defines a cohomology class $[c^{n+1}(f)] \in H^{n+1}(K; \pi_n(Y))$. We call $[c^{n+1}(f)]$ the $(n+1)$ -**th obstruction class** of f .

A similar concept is defined for homotopies. Suppose we have two maps $f_0, f_1 : K^{(n)} \rightarrow Y$ and a homotopy $h : K^{(n-1)} \times I \rightarrow Y$ between $f_0|_{K^{(n-1)}}$ and $f_1|_{K^{(n-1)}}$. We want to extend h to a homotopy $H : K^{(n)} \times I \rightarrow Y$ between f_0 and f_1 . To do this, we only need to extend h to each n -cell e_α^n . Let $\varphi_\alpha : S^{n-1} \rightarrow K^{(n-1)}$ be the attaching map of e_α^n , $\psi_\alpha : D^n \rightarrow K^{(n)}$ be the characteristic map of e_α^n . Then we can form a map

$$\begin{aligned} h_\alpha : (D^n \times \{0,1\}) \cup (\partial D^n \times I) &\rightarrow Y \\ (x, 0) &\mapsto f_0 \circ \psi_\alpha(x) \\ (x, 1) &\mapsto f_1 \circ \psi_\alpha(x) \\ (x, t) &\mapsto h \circ (\varphi_\alpha(x), t). \end{aligned}$$

Since $(D^n \times \{0,1\}) \cup (\partial D^n \times I) \cong S^n$, h_α defines a map from S^n to Y . It can extend to $D^n \times I \cong D^{n+1}$ if and only if $[h_\alpha] = 0$ in $\pi_n(Y)$. We in fact get a cochain $c^n(f_0, f_1, h) \in C_{cell}^n(K; \pi_n(Y))$:

$$\begin{aligned} c^n(f_0, f_1, h) : C_n^{cell}(K) &\rightarrow \pi_n(Y) \\ e_\alpha^n &\mapsto [h_\alpha]. \end{aligned}$$

We call $c^n(f_0, f_1, h)$ the **n -th deformation cochain** of (f_0, f_1, h) . If $f_0|_{K^{(n-1)}} = f_1|_{K^{(n-1)}}$ and h is the trivial homotopy, we simply write $c^n(f_0, f_1)$ for $c^n(f_0, f_1, h)$ and call it the **n -th difference cochain** of (f_0, f_1) . There are some basic facts about $c^n(f_0, f_1, h)$:

- h can be extended to a homotopy $H : K^{(n)} \times I \rightarrow Y$ between f_0 and f_1 if and only if $c^n(f_0, f_1, h) = 0$;
- (Naturality) Suppose L is another CW complex and $\phi : L \rightarrow K$ is a cellular map, then $c^n(f_0 \circ \phi, f_1 \circ \phi, h \circ (\phi \times \text{id}_I)) = \phi^\# c^n(f_0, f_1, h)$.
- $c^n(f_0, f_2, h * h') = c^n(f_0, f_1, h) + c^n(f_1, f_2, h')$, where $h * h'$ is the concatenation of h and h' . Especially, if $f_0|_{K^{(n-1)}} = f_1|_{K^{(n-1)}} = f_2|_{K^{(n-1)}}$ and h, h' are trivial homotopies, then $c^n(f_0, f_2) = c^n(f_0, f_1) + c^n(f_1, f_2)$.

In general, $c^n(f_0, f_1, h)$ is not a cocycle. But we have the following formula.

Theorem 1.2. $\delta c^n(f_0, f_1, h) = c^{n+1}(f_1) - c^{n+1}(f_0)$.

There is a useful lemma about difference cochains.

Lemma 1.3. *For any $f : K^{(n)} \rightarrow Y$ and any cochain $d \in C_{cell}^n(K; \pi_n(Y))$, there exists a map $f' : K^{(n)} \rightarrow Y$ such that $f'|_{K^{(n-1)}} = f|_{K^{(n-1)}}$ and $d^n(f, f') = d$.*

With the above preparations, we have the following theorem:

Theorem 1.4 (Eilenberg). *Given a map $f : K^{(n)} \rightarrow Y$, $[c^n(f)] = 0$ in $H^{n+1}(K; \pi_n(Y))$ if and only if f can extend to $K^{(n+1)}$ after having been changed on $K^{(n)} \setminus K^{(n-1)}$.*

Proof. $(\Leftarrow)$: If $\exists f' : K^{(n+1)} \rightarrow Y$ such that $f'|_{K^{(n-1)}} = f|_{K^{(n-1)}}$, then

$$c^{n+1}(f) = c^{n+1}(f) - c^{n+1}(f') = \delta d^n(f', f)$$

Thus $c^{n+1}(f)$ is a coboundary.

$(\Rightarrow)$: If $[c^{n+1}(f)] = 0$, then $\exists d \in C^n(K; \pi_n(Y))$ such that $c^{n+1}(f) = \delta d$. By lemma, there exists $f' : K^{(n)} \rightarrow Y$ such that $f|_{K^{(n-1)}} = f'|_{K^{(n-1)}}$ and $d^n(f, f') = -d$. So

$$c^{n+1}(f') = c^{n+1}(f) + \delta d^n(f, f') = \delta d - \delta d = 0.$$

Thus f' can be extended to $K^{(n+1)}$. □

Example 1.5. If the target space Y is n -connected, i.e., $\pi_k(Y) = 0$, $k = 0, 1, \dots, n$, then every map $f : K^{(0)} \rightarrow Y$ can be extended to $K^{(n+1)}$. Besides, any two maps $f, g : K^{(n)} \rightarrow Y$ are homotopic. Indeed, $f|_{K^{(0)}} \simeq g|_{K^{(0)}}$ for the reason that K is path-connected. We denote the homotopy by h_0 . Then h_0 can be extened to $h_n : K^{(n)} \times I \rightarrow Y$ since all the deformation cochain vanish.

Corollary 1.6. *If Y is n -connected and $\dim K \leq n$, then all the maps from K to Y are null-homotopic.*

Theorem 1.7 (Hopf-Whitehead). *For any CW complex K with $\dim K \leq n$ and $(n-1)$ -connected space Y , there is a one-to-one correspondence:*

$$[K, Y] \longleftrightarrow H^n(K; \pi_n(Y)).$$

If we weaken the dimensional condition and instead strengthen the target space to be an Eilenberg-MacLane space $K(\pi, n)$. Then we will get another theorem:

Theorem 1.8. *For any CW complex X , abelian group π and integer n , there is a one-to-one correspondence:*

$$[X, K(\pi, n)] \longleftrightarrow H^n(X; \pi).$$

Remark. In fact, there is a special class α in

$$H^n(K(\pi, n); \pi) \cong \text{Hom}(H_n(K(\pi, n)), \pi) \cong \text{Hom}(\pi, \pi)$$

which corresponds to the identity map in $\text{Hom}(\pi, \pi)$. The correspondence can be explicitly written by

$$\begin{aligned} [X, K(\pi, n)] &\rightarrow H^n(X; \pi) \\ f &\mapsto f^* \alpha. \end{aligned}$$

1.2 Characteristic Classes via Obstruction Theory

1.2.1 Cohomology with Local Coefficients

To describe characteristic classes as obstruction classes, we need cohomology with local coefficients.

Local Coefficient System:

Suppose X is a path-connected space. For any $x \in X$, let G_x be an abelian group attaching to x . For any path γ_{xy} from x to y , we have an isomorphism $(\gamma_{xy})_* : G_x \rightarrow G_y$. If $(\gamma_{xy})_*$ only depends on the homotopy class of γ rel. endpoints and satisfies the following conditions:

$$(\text{constant path})_* = \text{Id}, \quad (\gamma_{xy} * \gamma_{yz})_* = (\gamma_{yz})_* \circ (\gamma_{xy})_*.$$

Then we call $\mathcal{G} = \{G_x\}_{x \in X}$ a **local coefficient system** on X . If we fix a base point $x_0 \in X$, then $\mathcal{G}$ is determined by the group $G = G_{x_0}$ and an action $\rho : \pi_1(X, x_0) \rightarrow \text{Aut}(G)$. We call G the **typical group** of $\mathcal{G}$.

Example 1.9. If $\mathcal{G}$ is a local coefficient system on X with typical group G and trivial action. Then it is the ordinary constant coefficient we are familiar with.

Example 1.10. If $G_x = \mathbb{Z}_2$ for all $x \in X$, then since $\text{Aut}(\mathbb{Z}_2) = \{\text{Id}\}$, the local coefficient system with typical group $\mathbb{Z}_2$ must be trivial.

Example 1.11. If $G_x = \mathbb{Z}$ for all $x \in X$ and $\pi_1(X, x_0)$ acts on $\mathbb{Z}$ by multiplication of ± 1 according to whether the orientation is preserved or reversed along the loop. Then $\mathcal{G}$ is called the **orientation system** on X and denoted by Or .

Example 1.12. Let $p : \tilde{X} \rightarrow X$ be the universal covering of X . Then $\mathcal{G} = \{H_n(p^{-1}(x))\}_{x \in X}$ is a local coefficient system on X with typical group $H_n(\tilde{X})$ and action induced by the deck transformation group $\pi_1(X, x_0)$.

Example 1.13. Let $E \rightarrow X$ be a vector bundle with fiber F . Then $\mathcal{G} = \{H_n(p^{-1}(x))\}_{x \in X}$ is a local coefficient system on X with typical group $H_n(F)$ and action induced by the structure group of E .

Example 1.14. $\mathcal{G} = \{\pi_n(X, x)\}_{x \in X}$ is a local coefficient system on X with typical group $\pi_n(X, x_0)$ and action induced by the fundamental group $\pi_1(X, x_0)$.

Cohomology with Local Coefficients:

Let X be a CW complex and $\mathcal{G} = \{G_x\}_{x \in X}$ be a local coefficient system on X . We define the cellular cochain groups with local coefficients by

$$C_{cell}^n(X; \mathcal{G}) = \left\{ f : \{n\text{-cells of } X\} \rightarrow \bigsqcup_{x \in X} G_x \mid f(e_\alpha^n) \in G_{x_\alpha} \right\}$$

where x_α is a chosen point in the n -cell e_α^n . The coboundary operator $\delta : C_{cell}^n(X; \mathcal{G}) \rightarrow C_{cell}^{n+1}(X; \mathcal{G})$ is defined by

$$(\delta f)(e_\beta^{n+1}) = \sum_{\alpha} d_{\beta\alpha}(\gamma_{\alpha\beta})_* f(e_\alpha^n),$$

where $d_{\beta\alpha}$ is the degree of the map $\partial e_\beta^{n+1} \rightarrow X^{(n)} / (X^{(n)} - e_\alpha^n) \cong S^n$ and $\gamma_{\alpha\beta}$ is a path from x_α to x_β . It can be verified that $\delta^2 = 0$. Thus we can define the cohomology groups with local coefficients by

$$H^n(X; \mathcal{G}) = \frac{\ker(\delta : C_{cell}^n(X; \mathcal{G}) \rightarrow C_{cell}^{n+1}(X; \mathcal{G}))}{\text{im}(\delta : C_{cell}^{n-1}(X; \mathcal{G}) \rightarrow C_{cell}^n(X; \mathcal{G}))}.$$

The Homotopy Groups of $V_k(\mathbb{R}^n)$

Fact: The Stiefel manifold $V_k(\mathbb{R}^n)$ is $(n - k - 1)$ -connected and the first nontrivial homotopy group is

$$\pi_{n-k}(V_k(\mathbb{R}^n)) = \begin{cases} \mathbb{Z}, & n - k \text{ is even or } k = 1; \\ \mathbb{Z}_2, & n - k \text{ is odd and } k > 1. \end{cases}$$

Remark. It is evident that $V_k(\mathbb{R}^n)$ is a simple space.

Characteristic Classes

Now let's back to vector bundles. Let ξ be an $\mathbb{R}^n$ -bundle over a CW complex B . We want to investigate how nontrivial ξ is. To do this, we try to find k -linearly independent sections of ξ .

Let $V_k(\xi)$ be the Stiefel manifold bundle associated to ξ . It's a fiber bundle over B with fiber $F = V_k(\mathbb{R}^n)$. A section s of $V_k(\xi)$ is a k -frame of ξ . Locally on a trivializing neighborhood U_α , s can be identified as a map from U_α to $V_k(\mathbb{R}^n)$. Thus we can use obstruction theory to study the existence of sections of $V_k(\xi)$.

The obstructions $c^{j+1}(s)$ lie in $H^{j+1}(B; \mathcal{G})$, where $\mathcal{G}$ is a local coefficient system on B with typical group $\pi_j(V_k(\mathbb{R}^n))$. This is because although the restriction of $V_k(\xi)$ on every j -cell e_α^j is trivial, the transition function may not be identity, from which induces a nontrivial isomorphism of homotopy groups. Moreover, any path γ_{xy} from x to y can lift to a homotopy from F_x to F_y , thus induces an isomorphism

$$(\gamma_{xy})_* : \pi_j(F_x, x) \xrightarrow{\cong} \pi_j(F_y, y).$$

The theorems about obstruction classes can all generalize to cohomology with local coefficients without difficulty.

Now since $V_k(\mathbb{R}^n)$ is $(n-k-1)$ -connected, all the section s can be extended to $B^{(n-k)}$, all the sections on $B^{(n-k)}$ are homotopic with each other. The first obstruction comes when extend s to $(n-k+1)$ -skeleton, which we denote by

$$\mathfrak{o}_{n-k+1} := c^{n-k+1}(s) \in H^{n-k+1}\left(B; \{\pi_{n-k}(V_k(\mathbb{R}^n))\}\right)$$

Since all sections on $B^{(n-k)}$ are homotopic with each other, this obstruction class does not depend on the choice of s , i.e, $\mathfrak{o}_{n-k+1}$ it is well-defined.

Now we construct a series of characteristic classes from obstruction classes.

- If $n-k = 2j-1$ is odd and $1 < k < n$, then $\pi_{2j-1}(V_{n-2j+1}(\mathbb{R}^n)) = \mathbb{Z}_2$. In this case,

$$H^{2j}\left(B; \{\pi_{2j-1}(V_{n-2j+1}(\mathbb{R}^n))\}\right) \cong H^{2j}(B; \mathbb{Z}_2)$$

$$\mathfrak{o}_{2j} \mapsto w_{2j}(\xi).$$

- If $n-k = 2j$ is even and $1 < k < n$, then $\pi_{2j}(V_{n-2j}(\mathbb{R}^n)) = \mathbb{Z}$. After applying the mod 2 reduction, we have

$$H^{2j+1}\left(B; \{\pi_{2j}(V_{n-2j}(\mathbb{R}^n))\}\right) \xrightarrow{\text{mod } 2} H^{2j+1}(B; \mathbb{Z}_2)$$

$$\mathfrak{o}_{2j+1} \mapsto w_{2j+1}(\xi).$$

- If $k = n$, then $V_n(\mathbb{R}^n) = O(n)$, but $\pi_0(O(n)) = \mathbb{Z}_2$ is not a group. We turn to use $\tilde{H}^0(O(n); \mathbb{Z}) = \mathbb{Z}$ instead. Then $\tilde{H}^0(O(n); \mathbb{Z})$ is a local

coefficient system on B with typical group $\mathbb{Z}$ and action induced by the determinant representation $\det : \pi_1(B) \rightarrow O(n) \rightarrow \{\pm 1\}$. After applying the mod 2 reduction, we have

$$H^1\left(B; \{\tilde{H}^0(O(n))\}\right) \xrightarrow{\text{mod } 2} H^1(B; \mathbb{Z}_2)$$

$$\mathfrak{o}_1 \mapsto w_1(\xi).$$

- If $k = 1$, then $\pi_{n-1}(V_1(\mathbb{R}^n)) = \pi_{n-1}(S^{n-1}) = \mathbb{Z}$. After applying the mod 2 reduction, we have

$$H^n\left(B; \{\pi_{n-1}(V_1(\mathbb{R}^n))\}\right) \xrightarrow{\text{mod } 2} H^n(B; \mathbb{Z}_2)$$

$$\mathfrak{o}_n \mapsto w_n(\xi).$$

Cases	Obstruction classes	Characteristic classes
$n - k = 2j - 1, 1 < k < n$	$\mathfrak{o}_{2j} = w_{2j}$	
$n - k = 2j, 1 < k < n$	$\mathfrak{o}_{2j+1} \xrightarrow{\text{mod } 2} w_{2j+1}$	
$k = n$	$\mathfrak{o}_1 \xrightarrow{\text{mod } 2} w_1$	
$k = 1$	$\mathfrak{o}_n \xrightarrow{\text{mod } 2} w_n$	

Properties of our construction

Stiefel-Whitney Classes

Theorem 1.15. $w_i(\xi) \in H^i(B; \mathbb{Z}_2)$ are the Stiefel-Whitney classes of ξ .

Proof. omitted. □

$\mathfrak{o}_n$ and the Euler Class If ξ is orientable, then the structure group of ξ can be reduced to $SO(n)$. Thus the action of $\pi_1(B)$ on $\pi_{n-k}(V_k(\mathbb{R}^n))$ is trivial for $1 \leq k < n$. So the local coefficient system is trivial. In this case,

$$H^n\left(B; \{\pi_{n-1}(V_1(\mathbb{R}^n))\}\right) \cong H^n(B; \mathbb{Z}).$$

Theorem 1.16. If ξ is orientable, then $\mathfrak{o}_n \in H^n(B; \mathbb{Z})$ is the Euler class $e(\xi)$.

Proof. omitted. □

w_1 and Orientability The structure group of ξ can be reduced to $SO(n)$ if and only if the transition functions all have determinant 1, i.e., the action of $\pi_1(B)$ on $\tilde{H}^0(O(n); \mathbb{Z})$ is trivial. This is equivalent to the triviality of $\mathfrak{o}_1$. Thus

Proposition 1.17. $\mathfrak{o}_1 = 0$ if and only if ξ is orientable.

Theorem 1.18. $\mathfrak{o}_1$ has order 2, i.e., $2\mathfrak{o}_1 = 0$. Thus

$$w_1(\xi) = 0 \text{ if and only if } \mathfrak{o}_1 = 0.$$

Proof. The connected components of the space $V_n(\mathbb{R}^n)$ are identified with orientation of the fiber, and hence the elements of $\tilde{H}^0(V_n(\mathbb{R}^n); \mathbb{Z}) \cong \mathbb{Z}$ has the form $a\text{Or}_1 - a\text{Or}_2$; to each orientation we assign its Coefficient.

...

□

Is w_i the true obstruction? The fibration

$$S^{2j-1} \rightarrow V_{n-2j+1}(\mathbb{R}^n) \rightarrow V_{n-2j}(\mathbb{R}^n).$$

induces an exact sequence of homotopy groups

$$\pi_{2j}(V_{n-2j}(\mathbb{R}^n)) \xrightarrow{\partial} \pi_{2j-1}(S^{2j-1}) \rightarrow \pi_{2j-1}(V_{n-2j+1}(\mathbb{R}^n)) \rightarrow 0.$$

It is actually

$$\mathbb{Z} \xrightarrow{\partial} \mathbb{Z} \xrightarrow{i_*} \mathbb{Z}_2 \rightarrow 0.$$

Therefore $\text{Im } \partial = \text{Ker } i_* = 2\mathbb{Z}$ and $\text{Ker } \partial = 0$. So ∂ is multiplication by 2. So the exact sequence is a short exact sequence

$$0 \rightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \xrightarrow{i_*} \mathbb{Z}_2 \rightarrow 0.$$

This short exact sequence induces a long exact sequence of cohomology groups

$$\cdots \rightarrow H^{2j}(B; \mathbb{Z}) \xrightarrow{i_*} H^{2j}(B; \mathbb{Z}_2) \xrightarrow{\beta} H^{2j+1}(B; \mathbb{Z}) \xrightarrow{\times 2} H^{2j+1}(B; \mathbb{Z}) \rightarrow \cdots$$

where β is the Bockstein homomorphism.

Theorem 1.19. $\mathfrak{o}_{2j+1} = \beta(\mathfrak{o}_{2j})$.

Proof. omitted.

□

Corollary 1.20. $\mathfrak{o}_{2j+1}$ has order 2, i.e., $2\mathfrak{o}_{2j+1} = 0$. Thus

$$w_{2j+1}(\xi) = 0 \text{ if and only if } \mathfrak{o}_{2j+1} = 0.$$

In summary, almost all the Stiefel-Whitney classes w_i are the true obstructions, except when $i = n$ is even. If in addition ξ is oriented, then $\mathfrak{o}_n$ is Euler class $e(\xi)$.

2 Chern Numbers & Pontrjagin Numbers

2.1 Chern Numbers

Definition 2.1. Let K^n be a compact complex manifold, $\dim_{\mathbb{C}} K = n$. For each partition $I = i_1, \dots, i_r$ of n , we define the I -th Chern number to be the integer

$$c_I[K^n] = c_{i_1} \cdots c_{i_r}[K^n] := \langle c_{i_1}(\tau) \cdots c_{i_r}(\tau), \mu_{K^n} \rangle,$$

where μ_{K^n} is the fundamental homology class w.r.t. the orientation determined by the complex structure.

If I is a partition of other number, we define $c_I[K^n]$ to be 0.

Example 2.2. For $K^n = \mathbb{C}P^n$, we have

$$\begin{aligned} c_I[\mathbb{C}P^n] &= \langle c_{i_1}(\tau) \cdots c_{i_r}(\tau), \mu_{\mathbb{C}P^n} \rangle \\ &= \left\langle \binom{n+1}{i_1} \cdots \binom{n+1}{i_r} a^n, \mu_{\mathbb{C}P^n} \right\rangle \\ &= \binom{n+1}{i_1} \cdots \binom{n+1}{i_r}. \end{aligned}$$

2.2 Pontrjagin Numbers

Definition 2.3. Let M^{4n} be a compact oriented smooth manifold of dimension $4n$. For each partition $I = i_1, \dots, i_r$ of n , we define the I -th Pontrjagin number to be the integer

$$p_I[M^{4n}] = p_{i_1} \cdots p_{i_r}[M^{4n}] := \langle p_{i_1}(\tau) \cdots p_{i_r}(\tau), \mu_{M^{4n}} \rangle,$$

where $\mu_{M^{4n}}$ is the fundamental homology class w.r.t. the given orientation.

If I is a partition of other number, we define $p_I[M^{4n}]$ to be 0.

Example 2.4. For $M^{4n} = \mathbb{C}P^{2n}$, we have

$$\begin{aligned} p_I[\mathbb{C}P^{2n}] &= \langle p_{i_1}(\tau) \cdots p_{i_r}(\tau), \mu_{\mathbb{C}P^{2n}} \rangle \\ &= \left\langle \binom{2n+1}{i_1} \cdots \binom{2n+1}{i_r} a^{2n}, \mu_{\mathbb{C}P^{2n}} \right\rangle \\ &= \binom{2n+1}{i_1} \cdots \binom{2n+1}{i_r}. \end{aligned}$$

Oriented reversing map. If we reverse the orientation of M^{4n} , since the definition of Pontrjagin classes do not depend on the orientation, we have the following

$$p_k(M) \mapsto p_k(M),$$

$$\begin{aligned}\mu_M &\mapsto -\mu_M, \\ p_I[M] &\mapsto -p_I[M].\end{aligned}$$

Thus the sign of Pontrjagin numbers change when reversing the orientation of M . If M has a nonzero Pontrjagin number, M cannot possess any orientation reversing diffeomorphism.

Manifold as a boundary. If M^{4n} is a boundary of some smooth, compact, oriented $(4n + 1)$ -dimensional manifold-with-boundary. Then all Pontrjagin numbers of M must be zero. Thus if some $p_I[M] \neq 0$, then M cannot be the boundary of such manifold.

2.3 Symmetric Functions

We know that all symmetric polynomials in $\mathbb{Z}[t_1, \dots, t_n]$ is itself a polynomial ring:

$$\mathcal{S} = \mathbb{Z}[\sigma_1, \dots, \sigma_n] \subset \mathbb{Z}[t_1, \dots, t_n],$$

where σ_i is the i -th elementary symmetric polynomial.

$$1 + \sigma_1 + \dots + \sigma_n = (1 + t_1) \cdots (1 + t_n).$$

If we focus on the symmetric polynomials of degree k , denoted by $\mathcal{S}^k$, clearly it is a free abelian group with basis

$$\sigma_{i_1} \cdots \sigma_{i_r}, \quad I = i_1, \dots, i_r \text{ is a partition of } k.$$

For some reason, it is more convenient to use another basis of $\mathcal{S}^k$, which is described as follows.

Define two monomials in $\mathbb{Z}[t_1, \dots, t_n]$ to be equivalent if they differ by a permutation of the variables. We define

$$\sum t_1^{a_1} \cdots t_r^{a_r}$$

to be the summation of all distinct monomials equivalent to $t_1^{a_1} \cdots t_n^{a_n}$. Clearly, it is a symmetric polynomial.

Lemma 2.5.

$$\sum t_1^{a_1} \cdots t_r^{a_r}$$

where $a_1 \cdots a_r$ ranges over all partitions of k with length $r \leq n$, form another basis of $\mathcal{S}^k$.

Now for each partition $I = i_1, \dots, i_r$ of k , we define a polynomial s_I in k variables as follows: choose $n \geq k$, define

$$s_I(\sigma_1, \dots, \sigma_k) := \sum t_1^{i_1} \cdots t_r^{i_r},$$

both sides are symmetric polynomials in n variables.

Notes that s_I does not depend on the choice of n , for if we choose another $n' > n$, then we can set $t_{n+1} = \dots = t_{n'} = 0$ to get back to the previous case.

Clearly, all s_I form a basis of $\mathcal{S}^k$ as I ranges over all partitions of k .

Now let's look at the cohomology ring $H^*(\mathrm{Gr}_n(\mathbb{C}^\infty); \mathbb{Z})$. the splitting map

$$\phi : \mathbb{C}P^\infty \times \cdots \times \mathbb{C}P^\infty \rightarrow \mathrm{Gr}_n(\mathbb{C}^\infty)$$

induces an injective homomorphism of cohomology rings

$$\phi^* : H^*(\mathrm{Gr}_n(\mathbb{C}^\infty); \mathbb{Z}) \rightarrow H^*(\mathbb{C}P^\infty \times \cdots \times \mathbb{C}P^\infty; \mathbb{Z}) \cong \mathbb{Z}[t_1, \dots, t_n],$$

where t_i is the generator of the i -th factor. ϕ^* maps $H^*(\mathrm{Gr}_n(\mathbb{C}^\infty); \mathbb{Z}) = \mathbb{Z}[c_1, \dots, c_n]$ isomorphically onto the symmetric polynomial subring $\mathcal{S} \subset \mathbb{Z}[t_1, \dots, t_n]$, with c_i mapped to σ_i .

Since $\phi^* s_I(c_1, \dots, c_k) = s_I(\sigma_1, \dots, \sigma_k)$ forms a basis of $\mathcal{S}^k$ as I ranges over all partitions of k , by the injectivity of ϕ^* , $s_I(c_1, \dots, c_k)$ also forms a basis of $H^{2k}(\mathrm{Gr}_n(\mathbb{C}^\infty); \mathbb{Z})$.

2.4 A Product Formula

Let ω be an n -dimensional complex vector bundle over a compact complex manifold B . For each $k \geq 0$ and partition $I = i_1, \dots, i_r$ of k , we define

$$s_I(c(\omega)) := s_I(c_1(\omega), \dots, c_k(\omega)) \in H^{2k}(B; \mathbb{Z}).$$

sometimes we simply write it as $s_I(c)$.

Lemma 2.6 (Thom).

$$s_I(c(\omega \oplus \omega')) = \sum_{J, K} s_J(c(\omega)) s_K(c(\omega')),$$

where the summation is taken over all partitions J, K with juxtaposition JK equal to I .

Remark. We can view this formula as a generalization of the Whitney sum formula.

Proof. Let $\phi : F(B) \rightarrow B$ be the flag manifold of $\omega \oplus \omega'$, with the splitting bundle

$$\phi^*(\omega) = \eta_1 \oplus \cdots \oplus \eta_n, \quad \phi^*(\omega') = \eta'_1 \oplus \cdots \oplus \eta'_{n'}.$$

Then we have

$$\begin{aligned} \phi^*c(\omega) &= (1 + c_1(\eta_1)) \cdots (1 + c_1(\eta_n)) =: (1 + t_1) \cdots (1 + t_n), \\ \phi^*c(\omega') &= (1 + c_1(\eta'_1)) \cdots (1 + c_1(\eta'_{n'})) =: (1 + t_{n+1}) \cdots (1 + t_{n+n'}). \end{aligned}$$

Thus

$$\begin{aligned} \phi^*s_I(c(\omega \oplus \omega')) &= \sum t_1^{i_1} \cdots t_r^{i_r} \\ &= \sum_{\substack{JK=I, \\ J=j_1, \dots, j_{r_1}, \\ K=k_1, \dots, k_{r_2}}} \left(\sum t_1^{j_1} \cdots t_{r_1}^{j_{r_1}} \right) \left(\sum t_{n+1}^{k_1} \cdots t_{n+r_2}^{k_{r_2}} \right) \\ &= \sum_{JK=I} s_J(c(\eta_1 \oplus \cdots \oplus \eta_n)) s_K(c(\eta'_1 \oplus \cdots \oplus \eta'_{n'})) \\ &= \sum_{JK=I} s_J(\phi^*c(\omega)) s_K(\phi^*c(\omega')) \\ &= \phi^* \left(\sum_{JK=I} s_J(c(\omega)) s_K(c(\omega')) \right). \end{aligned}$$

Since ϕ^* is injective, we get the desired formula. $\square$

Now for a compact complex manifold K^n . For each partition $I = i_1, \dots, i_r$ of n , we denote

$$s_I[K^n] = s_I(c)[K^n] := \langle s_I(c(\tau)), \mu_{K^n} \rangle \in \mathbb{Z}.$$

This characteristic number is clearly a linear combination of Chern numbers.

Corollary 2.7.

$$s_I[K^m \times L^n] = \sum_{JK=I} s_J[K^m] s_K[L^n].$$

where the summation is taken over all partitions J of m and K of n with juxtaposition JK equal to I .

Proof. Since $\tau_{K \times L} = \tau_K \times \tau_L = \pi_1^* \tau_K \oplus \pi_2^* \tau_L$, where π_1, π_2 are the projections from $K \times L$ to K and L respectively,

$$s_I[K \times L] = \langle s_I(c(\tau_{K \times L})), \mu_{K \times L} \rangle$$

$$\begin{aligned}
&= \sum_{JK=I} \langle s_J(c(\pi_1^* \tau_K)) s_K(c(\pi_2^* \tau_L)), \mu_K \times \mu_L \rangle \\
&= \sum_{JK=I} \langle s_J(c(\tau_K)), \mu_K \rangle \langle s_K(c(\tau_L)), \mu_L \rangle \\
&= \sum_{JK=I} s_J[K] s_K[L].
\end{aligned}$$

□

Corollary 2.8. *As a special case, $s_{m+n}[K^m \times L^n] = 0$, for $m, n > 0$.*

There are analogous results for Pontrjagin classes and numbers. Let ξ be a real vector bundle over a compact smooth manifold B . For each $k \geq 0$ and partition $I = i_1, \dots, i_r$ of k , we define

$$s_I(p(\xi)) := s_I(p_1(\xi), \dots, p_k(\xi)) \in H^{4k}(B; \mathbb{Z}).$$

There are similar product formulas for Pontrjagin classes

$$s_I(p(\xi \oplus \xi')) \equiv \sum_{J,K} s_J(p(\xi)) s_K(p(\xi')),$$

modulo elements of 2-torsions. Since pairing with the fundamental class annihilates all 2-torsions, we have

$$s_I[M \times N] = \sum_{JK=I} s_J[M] s_K[N],$$

Notes that these characteristic numbers will be zero unless both the dimensions of M and N are divisible by 4.

2.5 Linear Independence of Chern Numbers and of Pontrjagin Numbers

Theorem 2.9 (Thom). *Let $K^1, \dots, K^n$ be compact complex manifolds with $s_k(c)[K^k] \neq 0$. Then the $p(n) \times p(n)$ matrix*

$$\left(c_{i_1} \cdots c_{i_r} [K^{j_1} \times \cdots \times K^{j_s}] \right)$$

is nonsingular, where both $I = i_1, \dots, i_r$ and $J = j_1, \dots, j_s$ range over all partitions of n .

Similarly, we have the following theorem for Pontrjagin numbers.

Theorem 2.10 (Thom). *Let $M^4, \dots, M^{4n}$ be compact oriented smooth manifolds with $s_k(p)[M^{4k}] \neq 0$. Then the $p(n) \times p(n)$ matrix*

$$\left(p_{i_1} \cdots p_{i_r} [M^{4j_1} \times \cdots \times M^{4j_s}] \right)$$

is nonsingular, where both $I = i_1, \dots, i_r$ and $J = j_1, \dots, j_s$ range over all partitions of n .

Proof. We only prove the first theorem, the second one is completely analogous.

Since the basis $c_{i_1} \cdots c_{i_r}$ isn't well behaved under the product operation, we change to the basis $s_I(c)$. By the product formula,

$$s_I[K^{j_1} \times \cdots \times K^{j_s}] \neq 0$$

only if $I = i_1, \dots, i_r$ is a refinement of $J = j_1, \dots, j_s$. Thus by suitable ordering of both the rows and columns, we can assume that the matrix is upper triangular. The diagonal elements are

$$s_{i_1, \dots, i_r} [K^{i_1} \times \cdots \times K^{i_r}] = s_{i_1} [K^{i_1}] \cdots s_{i_r} [K^{i_r}] \neq 0.$$

Thus the matrix is nonsingular. □

3 Thom Space & Transversality

Definition 3.1. Let ξ be a rank k vector bundle over a base space B with total space $E(\xi)$, suppose ξ is equipped with a Euclidean metric. Let

$$A = \{v \in E(\xi) \mid \|v\| > 1\}.$$

The **Thom space** of ξ is defined to be the quotient space

$$T(\xi) = E(\xi)/A.$$

Lemma 3.2. *If the base space B is a CW-complex, then the Thom space $T(\xi)$ is a $(k-1)$ -connected CW-complex, having one cell of dimension $n+k$ for each cell of dimension n in B and a base point t_0 .*

Remark. Notice that by the cellular approximation theorem, a CW-complex with k -skeleton trivial is $(k-1)$ -connected.

Lemma 3.3. *If ξ is an orientable rank k vector bundle over B , then*

$$H_{k+i}(T(\xi), t_0; \mathbb{Z}) \cong H_i(B; \mathbb{Z}).$$

Proof.

$$\begin{aligned} H_{k+i}(T(\xi), t_0; \mathbb{Z}) &= H_{k+i}(E(\xi)/A, E_0/A; \mathbb{Z}) \\ &\cong H_{k+i}(E(\xi), E_0; \mathbb{Z}) \quad \text{by excision thm} \\ &\cong H_i(B; \mathbb{Z}) \quad \text{by Thom isom.} \end{aligned}$$

□

Theorem 3.4. *Let X be a $(k-1)$ -connected finite CW-complex, $k \geq 2$. Then the Hurewicz homomorphism*

$$\pi_r(X) \rightarrow H_r(X; \mathbb{Z})$$

is a $\mathcal{C}$ -isomorphism for $r < 2k-1$.

Corollary 3.5. *If T is the Thom space of an **orientable** rank k vector bundle over a **finite** CW-complex B , $k \geq 2$, then there is a $\mathcal{C}$ -isomorphism*

$$\pi_{k+i}(T) \rightarrow H_i(B; \mathbb{Z})$$

for $i < k-1$.

Proof.

$$\pi_{k+i}(T) \stackrel{\mathcal{C}\text{-isom}}{\cong} H_{k+i}(T, t_0; \mathbb{Z}) \cong H_i(B; \mathbb{Z}), \quad i < k - 1.$$

The first isomorphism holds since by Lemma 3.2 T is $(k-1)$ -connected finite CW-complex, then we can apply Theorem 3.4; the second isomorphism holds since ξ is orientable, we can apply Lemma 3.3. $\square$

Theorem 3.6. *Every continuous map $f: S^m \rightarrow B$ is homotopic to a map g which is smooth throughout $g^{-1}(T - t_0)$ and transverse to the zero section B . The oriented cobordism class of the submanifold $g^{-1}(B)$ depends only on the homotopy class of g . Hence there is a well-defined map*

$$\begin{aligned} \pi_m(T, t_0) &\rightarrow \Omega_{m-k}^{\text{SO}} \\ [g] &\mapsto [g^{-1}(B)]. \end{aligned}$$

3.1 The Main Theorem

Let $\tilde{\gamma}^k$ be the universal oriented rank k vector bundle over $\tilde{\text{Gr}}_k(\mathbb{R}^\infty)$, we have the following important theorem.

Theorem 3.7 (Thom's Theorem). *For $k > n + 1$, the map*

$$\begin{aligned} \pi_{n+k}(T(\tilde{\gamma}^k), t_0) &\rightarrow \Omega_n^{\text{SO}} \\ [g] &\mapsto [g^{-1}(\tilde{\text{Gr}}_k(\mathbb{R}^\infty))] \end{aligned}$$

is an isomorphism. Similarly, for the unoriented case, the map

$$\begin{aligned} \pi_{n+k}(T(\tilde{\gamma}^k), t_0) &\rightarrow \Omega_n \\ [g] &\mapsto [g^{-1}(\tilde{\text{Gr}}_k(\mathbb{R}^\infty))] \end{aligned}$$

is also an isomorphism.

Remark. 这个同构是否由这个映射给出存疑.

We will not prove this theorem, instead, we will proof part of it. Let $\tilde{\gamma}_p^k$ be the universal bundle over $\tilde{\text{Gr}}_k(\mathbb{R}^{k+p})$.

Lemma 3.8. *If $k \geq n$ and $p \geq n$, then the homomorphism*

$$\begin{aligned} \pi_{n+k}(T(\tilde{\gamma}_p^k), t_0) &\rightarrow \Omega_n^{\text{SO}} \\ [g] &\mapsto [g^{-1}(\tilde{\text{Gr}}_k(\mathbb{R}^{k+p}))] \end{aligned}$$

is surjective.